RRT with Dense sampling is probabilistically complete

I Properties of Sequence

A) Nice property (for now)

1) Low dispersion

2) Place samples in a way that the largest uncovered area be a small as possible

3) Known as Voronoi Property

4) Great for exploration

II Store New Edge

A) Exact (Computationally Expensive)

1) 2D line segment

2) Configuration Space

3) State space

B) Approximate (Good for RRT but Bad for RRT*) — Also uses a lot of memory

1) When storing new edge, insert intermediate point

2) Finding nearest point in graph is just finding a vertex

C) Vertex Only (Bad for RRT, Great for RRT*)

1) Don't worry about actually finding closest point

2) Like (B) but no intermediate points

III Finding Closest Point for Approximate Solutions

Use a tree